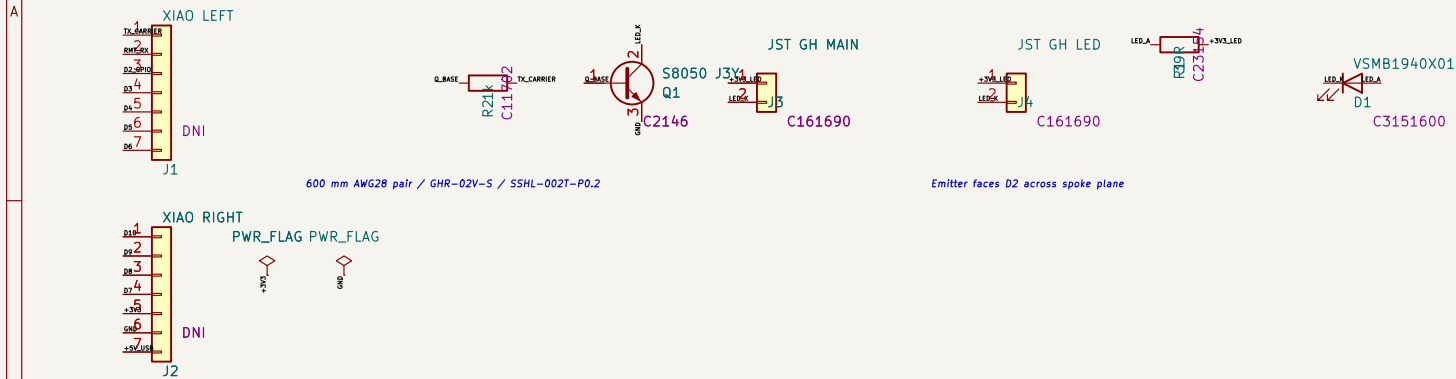


1 ESP32-S3 HOST / RMT

2 DRIVER / LOCKING HARNESS

3 REMOTE EMITTER PCB

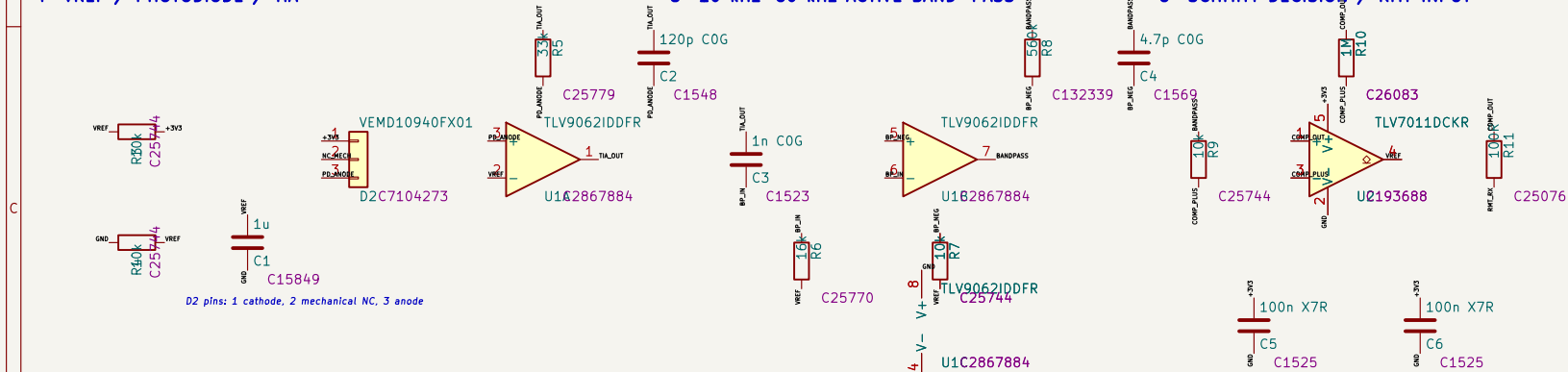


GPIO1: runtime carrier TX | GPIO2: comparator carrier into RMT RX

4 VREF / PHOTODIODE / TIA

5 10 kHz–60 kHz ACTIVE BAND-PASS

6 SCHMITT DECISION / RMT INPUT



36.9 mV hysteresis typ.; RMT RX removes selected carrier

SYSTEM NOTES

– RMT TX carrier is runtime-selectable from 25 kHz to 50 kHz. The discrete AFE rejects DC sunlight; the Schmitt stage restores logic. RMT RX carrier removal yields blockage intervals for the adaptive LUT. J3/J4 are the harness boundary; J4/R1/D1 are on the remote PCB.

Sheet: /
File: ir_spoke_link.kicad_sch

Title: IR Spoke Sensor

Size: A4	
KiCad E.D.A. 10.0.5	

Date:

Rev:

Id: 1/1